BikeWorkshop

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Current DIY Bike Mechanics

Motivation

Information rich but
Does **NOT** Simulate
Actual Movements

Interactive, with information
and Simulate Actual
Movements

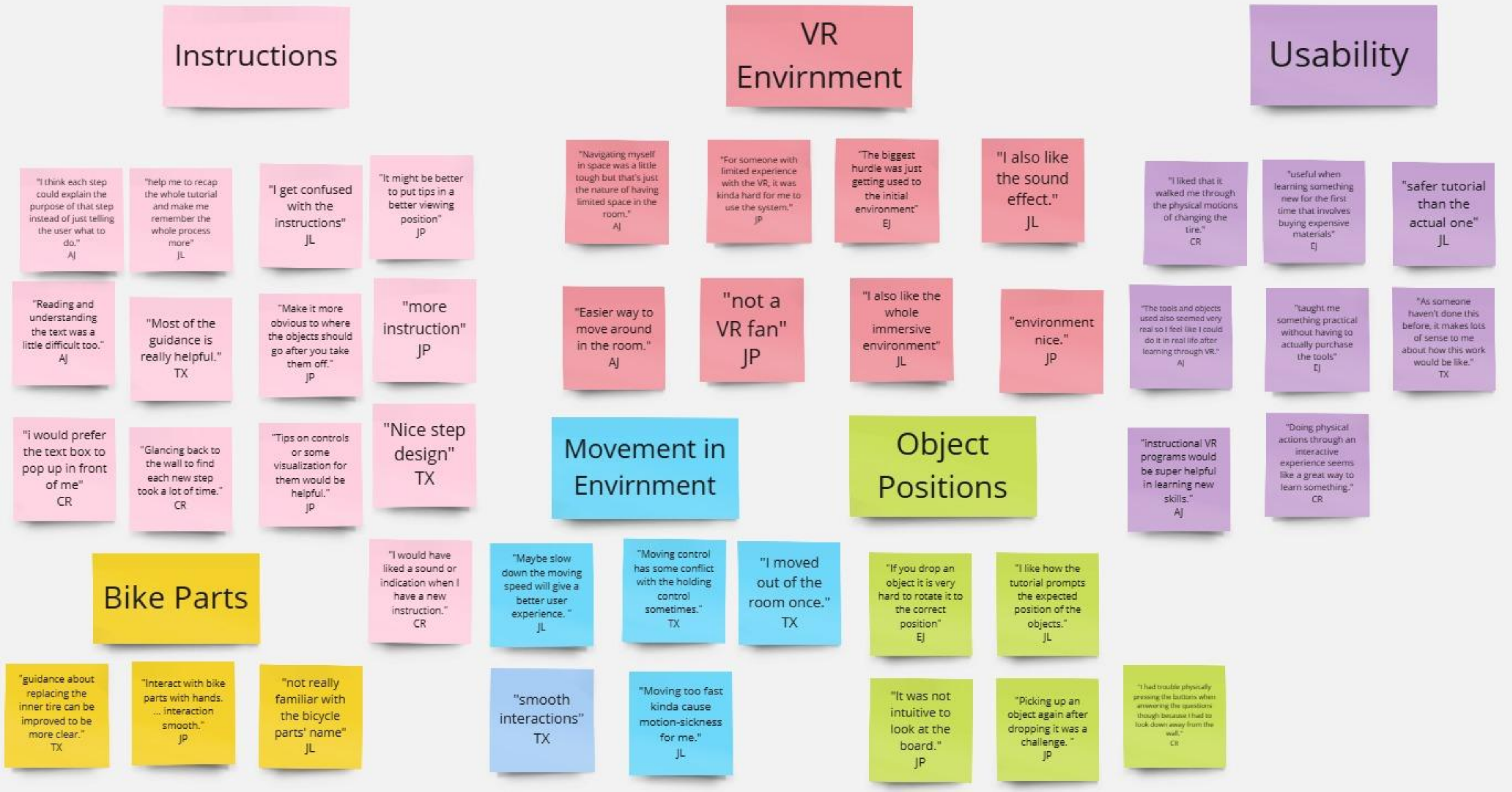
Novelty

- VR Learning is an engaging way to learn and be able to interact with the bike in a low consequence way, allowing you to master the process before attempting it on your own bike
- Do the movements of repairing a bike which are analogous to real life
- Bike maintenance can be expensive, but for simple fixes, can be fun and easy to learn

Impact

- In many places of the world bikes are the main mode of transportation, one way to learn to take care of your bike from an expert without having to physically be there

BikeWorkshop Walk-Through


Instructions
"I think each step could explain the purpose of that step instead of just telling the user what to do." AJ
"help me to recap the whole tutorial and make me remember the whole process more" JL
"I get confused with the instructions" JL
"It might be better to put tips in a better viewing position" JP
"Reading and understanding the text was a little difficult too." AJ
"Most of the guidance is really helpful." TX
"Make it more obvious to where the objects should go after you take them off." JP
"more instruction" JP
"i would prefer the text box to pop up in front of me" CR
"Glancing back to the wall to find each new step took a lot of time." CR
"Tips on controls or some visualization for them would be helpful." JP
"Nice step design" TX
"I would have liked a sound or indication when I have a new instruction." CR
Bike Parts
"guidance about replacing the inner tire can be improved to be more clear." TX
"Interact with bike parts with hands. ... interaction smooth." JP
"not really familiar with the bicycle parts' name" JL
VR Envirnment
"Navigating myself in space was a little tough but that's just the nature of having limited space in the room." AJ
"For someone with limited experience with the VR, it was kinda hard for me to use the system." JP
"The biggest hurdle was just getting used to the initial environment" EJ
"I also like the sound effect." JL
"Easier way to move around in the room." AJ
"not a VR fan" JP
"I also like the whole immersive environment" JL
"environment nice." JP
Movement in Envirnment
"Maybe slow down the moving speed will give a better user experience. " JL
"Moving control has some conflict with the holding control sometimes." TX
"I moved out of the room once." TX
"smooth interactions" TX
"Moving too fast kinda cause motion-sickness for me." JL
Object Positions
"If you drop an object it is very hard to rotate it to the correct position" EJ
"I like how the tutorial prompts the expected position of the objects." JL
"It was not intuitive to look at the board." JP
"Picking up an object again after dropping it was a challenge. " JP
"I had trouble physically pressing the buttons when answering the questions though because I had to look down away from the wall." CR
Usability
"I liked that it walked me through the physical motions of changing the tire." CR
"useful when learning something new for the first time that involves buying expensive materials" EJ
"safer tutorial than the actual one" JL
"The tools and objects used also seemed very real so I feel like I could do it in real life after learning through VR." AJ
"taught me something practical without having to actually purchase the tools" EJ
"As someone haven't done this before, it makes lots of sense to me about how this work would be like." TX
"instructional VR programs would be super helpful in learning new skills." AJ
"Doing physical actions through an interactive experience seems like a great way to learn something." CR

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User Study Results

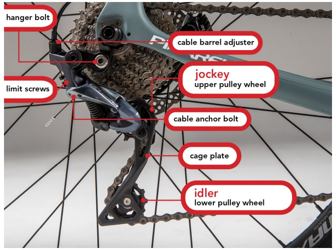
Positive Feedback:

- Immersiveness of the bike environment including the workshop and sound effects
- Practicality of concept with real world applications

What we can improve on in the next iteration:

- Dictionary for bike terms and parts (with photos/ demo parts)
- Smooth movement and make controls more intuitive
- Having animations for interactions to help guide the user
- Make objects more easily retrievable once dropped
- Creating bigger boundary boxes to reduce object dropping

Conclusion



Summary:

- BikeWorkshop is a tool to help bike owners learn how to fix their bikes without needing the proper tools or running the risk of damaging their bike
- This current rendition of the project runs through the tutorial of how to change a tire with added tips and quizzes to enhance the learning experience

Future:

- Updating the current project with User Study suggestions
- Additional tutorials for different bike types, fixing different parts of the bike (i.e. derailleur for shifting)
- Extending to AR environments to help answer your questions as they arise on your actual bike